

Core Commutative Diagrams for Coefficient-Split CKKS Bootstrapping

This note contains two commutative diagrams for the ideal normalized coefficient-split CKKS bootstrapping core. The first diagram isolates the key semantic invariant: coefficient split does not preserve the original big-ring canonical slots, but after applying C2S, the resulting coefficient-packed coordinates decompose by residue class. The second diagram shows the full split-compatible sandwich.

1 Core Split–C2S Commutative Square

The key compatibility is

$$\left(\bigoplus_{r=0}^{k-1} \text{C2S}_n \right) \circ \text{Split}_k = \Pi_k \circ \text{C2S}_N.$$

Equivalently, the following square commutes.

$$\begin{array}{ccc}
 \text{Can}_N & \xrightarrow{\text{Split}_k} & \bigoplus_{r=0}^{k-1} \text{Can}_n \\
 \downarrow \text{C2S}_N & & \downarrow \bigoplus_{r=0}^{k-1} \text{C2S}_n \\
 \text{Coeff}_N & \xrightarrow{\Pi_k} & \bigoplus_{r=0}^{k-1} \text{Coeff}_n
 \end{array}$$

Figure 1: Core semantic commutative square. The split operation does not preserve the original big-ring canonical slots. However, after applying C2S, the resulting coefficient-packed coordinates decompose exactly by residue class. This is the invariant used by the factorized bootstrapping core.

The top route first performs coefficient split at the message level and then applies leaf C2S_n. The bottom route first applies the large-ring C2S_N and then groups coefficient-packed coordinates by residue class using Π_k. The two routes produce the same leaf coefficient-packed representation.

2 Full Split-Compatible Sandwich

The ideal large-ring core is

$$\text{BTS}_N^{\text{core}} = \text{S2C}_N \circ \text{EvalMod}_N \circ \text{C2S}_N.$$

The leaf core is

$$\text{BTS}_n^{\text{core}} = \text{S2C}_n \circ \text{EvalMod}_n \circ \text{C2S}_n.$$

The factorization theorem states that, under the normalized coefficient-split semantics,

$$\text{S2C}_N \circ \text{EvalMod}_N \circ \text{C2S}_N = \text{Merge}_k \circ \left(\bigoplus_{r=0}^{k-1} \text{S2C}_n \circ \text{EvalMod}_n \circ \text{C2S}_n \right) \circ \text{Split}_k.$$

The proof is visualized by the following diagram.

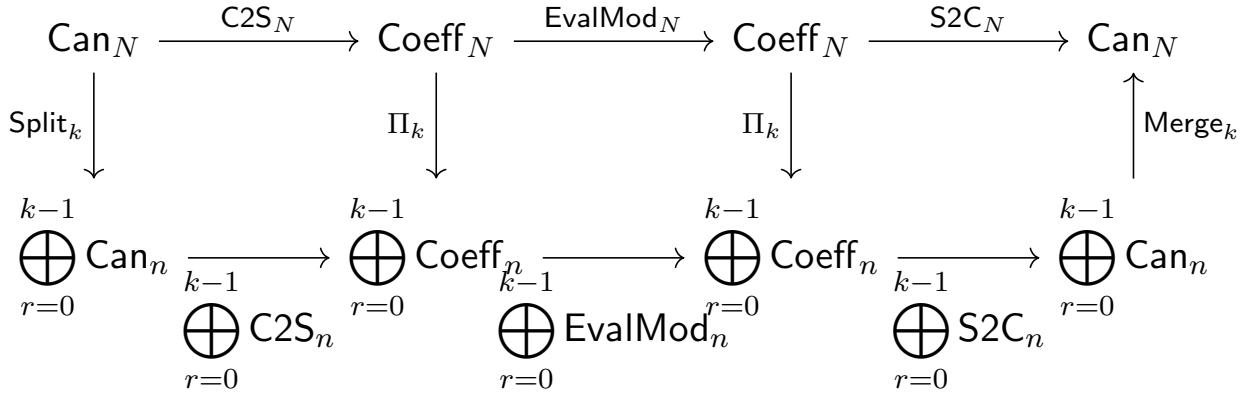


Figure 2: Factorized CKKS bootstrapping core as a split-compatible sandwich. The left square is the Split–C2S compatibility, the middle square is component-wise EvalMod compatibility, and the right rectangle is the S2C–Merge compatibility. Together, these compatibilities imply the factorization of the large-ring bootstrapping core into leaf cores.

The left square expresses

$$\left(\bigoplus_{r=0}^{k-1} \text{C2S}_n \right) \circ \text{Split}_k = \Pi_k \circ \text{C2S}_N.$$

The middle square expresses the component-wise EvalMod compatibility

$$\left(\bigoplus_{r=0}^{k-1} \text{EvalMod}_n \right) \circ \Pi_k = \Pi_k \circ \text{EvalMod}_N.$$

The right rectangle expresses

$$\text{Merge}_k \circ \left(\bigoplus_{r=0}^{k-1} \text{S2C}_n \right) \circ \Pi_k = \text{S2C}_N.$$

Composing these three compatibilities gives the factorized core identity.